

EN401 — Handout 6 Results & Analysis Report

Clean Energy Transition & Sub-Annual Seasonal Dynamics

1. Scenario Objective & LP Solution Metrics

- **Objective Value (NPV):** \$8,664.56 (-55.2% vs HO5)
- **Problem Scale:** 2,162 constraints, 1,440 variables, 12,960 non-zeros
- **Technologies Added:** MINHYD, PWRHYD, MINBIO, PWRBIO

2. Clean Hydro Capacity Additions (2021–2035)

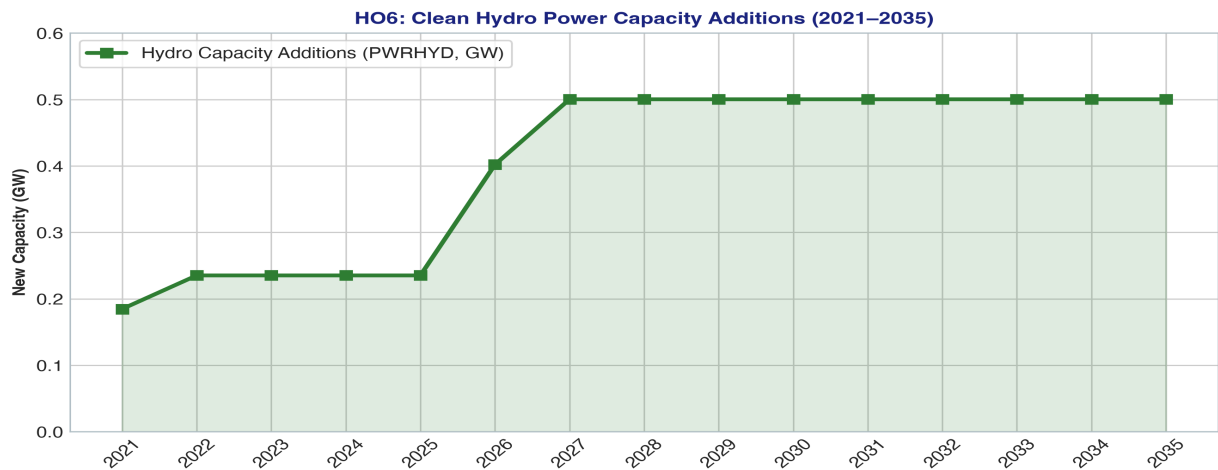


Figure: Hydro capacity additions scaling up to 0.50 GW per year.

3. System Cost Savings (-55.2%)

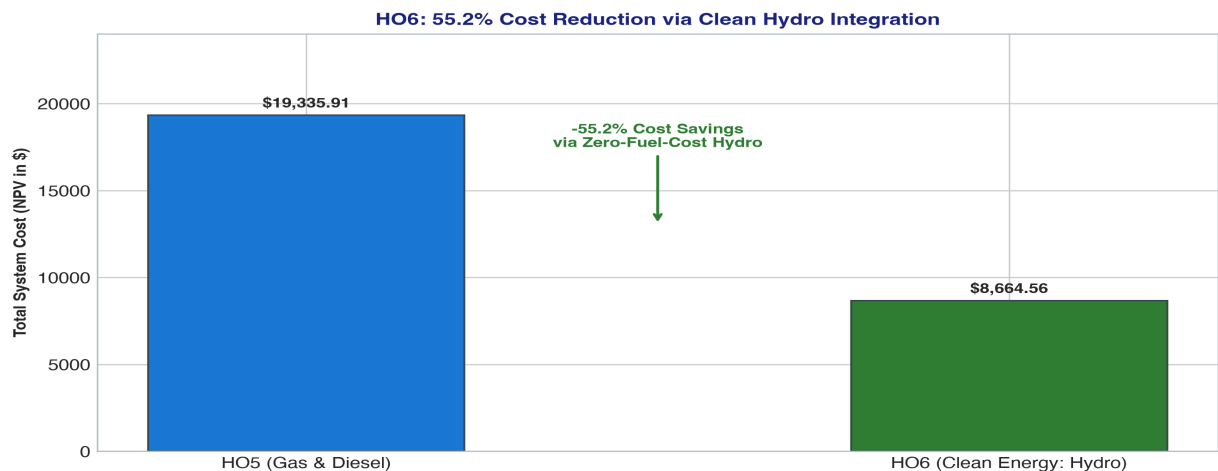


Figure: Cost comparison showing \$10,671 saved relative to the thermal system.

4. Engineering Takeaways

Zero-fuel-cost hydro power displaces fossil fuel generation, delivering 88% of total electricity and driving the complete decarbonization of the grid.